

Modern rammed earth construction techniques

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Abstract: This chapter provides an overview of modern rammed earth construction techniques, with a particular emphasis on current practices in the United States. It describes all of the key steps in constructing rammed earth wall systems, recognizing the myriad alternative approaches that exist for every part of the process. The chapter ends with a reflection on likely future trends and a discussion of factors that could contribute to growth in the field.

Key words: rammed earth, amending mix designs, machine mixing, manufactured form systems, conveyor delivery.

14.1 Introduction

The earliest examples of rammed earth were made by compacting one damp soil layer at a time into temporary wooden formwork. Soil could be dug by hand from the site and, when the moisture was suitable, installed directly into the formwork. Time, simple formwork and a compacting tool were essentially the only resources required for constructing these modest but long-lasting structures.

Over the course of centuries, advancements in the field have been significant. The aesthetic associated with modern rammed earth is hardly comparable to its humble ancestry (see Fig. 14.1); engineering specifications may require that rammed earth be stabilized with cement, reinforced with steel and designed to reach compressive strengths in excess of 3000 pounds per square inch (psi) (20.68 MPa). In its most complex applications, rammed earth is an engineered, high-strength, environmentally responsible building material installed with sophisticated machinery. The low-tech, labor-intensive version of rammed earth construction still has great value, however. In developing nations, labor is inexpensive and conventional building resources scarce. There, the less sophisticated method of building rammed earth can prove appropriate and effective.

Rammed earth construction has a wide range of approaches. In one scenario, an installation can involve a single form made of reuseable, hand-sawn lumber, shovels, buckets and heavy wooden rammers. Materials for



14.1 Rammed earth with modern aesthetic, Las Vegas, NV.

construction can be harvested locally, forming lumber can be reused for roofing or other construction applications, and unskilled volunteer labor can be utilized. The modern method involves laminate plywood, aluminum form supports, conveyor belts, air compressors, backfill tampers, tractors, mixing machines and laboratory testing.

In California, engineering requirements demand that rammed earth be stabilized with cement to improve compressive strength and other durability characteristics, reinforced with steel to protect wall systems from failure in the event of seismic activity and amended with imported aggregate materials to optimize quality. High labor rates require that many workers with non-mechanical tools be replaced with capital-intensive, labor-lean systems. Rammed earth's acceptance into the United States building industry, especially seismically active California, has naturally been biased towards reinforced and stabilized methods.

The resulting range of approaches to building with rammed earth can be highly variable in (i) cost, (ii) labor requirements, (iii) production times and (iv) associated ecological impacts.

Although techniques for the application of rammed earth vary considerably from region to region, the following chapter has been organized and written to provide a broad working knowledge of the various aspects of the process, from selection of raw materials to treatment of finished walls. It is heavily weighted towards the authors' direct experience – which has been primarily gained in California.

14.2 Material sourcing

The strength and look of a finished rammed earth wall is directly governed by the raw materials in the mix design, and material sourcing and processing have significant financial and environmental impacts on a project. The final selection of bulk materials will involve making trade-offs between cost of materials and wall quality. Geotechnical evaluation in conjunction with trial mix designs and pre-construction testing is highly recommended.

14.2.1 Site-sourced materials

In the ideal situation, suitable soil materials required for construction can be found on the building site, and required quantities can be matched to excavations associated with site work. On small projects, site sourcing will be limited to the footprint of the structure. On larger parcels it is often possible to encounter different soil profiles, one or more of which may have the required characteristics.

In many cases, before a site-sourced material can be used in a mix formulation, screening or other forms of processing may be required. For smaller projects, a screening operation may be as simple as shoveling raw material through a static screen. Larger projects requiring greater quantities may warrant front-loaders and vibratory screening plants. Our experience has indicated that the expenses associated with on-site processing can in some cases equal the price per yard of an imported quarry product.

14.2.2 Amending and blending

In most cases, a suitable mix design can be developed using some percentage of site-excavated soil in combination with one or more imported amendments. Optimum proportioning between site soil and amendment is determined through materials testing. Cost and intended wall quality will govern the decision. Imported materials can be expensive but their inclusion can enhance wall characteristics and reduce stabilization requirements. Common field and laboratory testing for rammed earth material inputs can be found in Chapter 6 of this text, as well as in Section 14.7 of this chapter.

Importable amendments typically come from quarries. Occasionally excess soil material from some close or adjacent construction project such as a pipeline, road cut, basement or pool excavation might be obtained for the cost of transport. Be aware that free imports may require on-site screening to reduce gravel size and to break-up clods to improve the materials' suitability.

The second source for amendment is virgin aggregate from a rock, sand or gravel quarry. The advantages to importing from a quarry are that the selected

material is usually uniform, consistent and requiring no on-site processing. Additionally, materials can be delivered in manageable, incremental quantities, which can be of benefit on small sites.

14.2.3 Quarry waste and recycled materials

Quarries may prove an even greater resource if they have what can be classified as post-industrial aggregates that are suitable for use as soil amendments. Most quarry operations are geared to the manufacture of clean building products with characteristics that adhere to industry standards. In many cases, the production of these materials results in the creation of lesser value by-products. These by-products may be 'overburden', 'crusher fines' or 'tailings' – essentially waste products generated during primary processing. These types of products are in many cases suitable for use in rammed earth mix designs, with the nature of their creation allowing them to be classified as post-industrial recycled. This can work to double advantage for the environmentally conscious rammed earth builder – they are sold at reduced price and can reduce the ecological impacts associated with construction. Furthermore, the LEED rating system awards points for the use of post-industrial recycled materials.

Recycled brick, tile and concrete can also act as amendments for a balanced rammed earth mix formulation. Gradation varies depending on the input and the processing. Advantages include (i) color enhancement: (brick and tile can introduce strong colors into the mix) and; (ii) increased strength (recycled concrete in some cases retains free cement, which can allow for a reduction in stabilization requirements).

14.2.4 Transportation expenses

Heavy trucks and diesel fuel account for a high percentage of the cost of soil amendment. It follows that the further the amendment source is from the project, the more significant are the environmental (and monetary) costs associated with the importation of that amendment. The choice of a final mix design, therefore, will involve decision-making in three key areas: (i) wall quality and performance; (ii) raw material costs and (iii) ecological impacts of construction. An argument might be made that the more durable the wall (i.e. improved wall quality) the more years of service it can provide, thus offsetting potentially higher initial construction costs (both monetary and ecological). Conversely, a rammed earth wall built entirely of site-sourced material with low to no stabilization will consume fewer non-renewable resources but may not have the same durability. Life-cycle costing takes into account both the energy inputs and the life expectancy of the finished product.

14.2.5 Stockpiling and moisture control

Site logistics are an important factor in staging any rammed earth construction project. Stockpiling raw materials, proportioning and blending, delivering the prepared mix to the formwork – all require substantial room on the site and around the building perimeter. When a portion of site-soil is to be used in the formulation, this will have been excavated in previous stages of construction and should be stored near the work area but out of the paths of other trades. Imported material must be off-loaded and stored so that it is both accessible and convenient. Work areas for carpenters to construct formwork must be maintained. Careful pre-construction planning during the early stages of construction can reduce wasted man-hours and equipment costs.

Once the selected soil and aggregate materials have been stockpiled on site, it is important to ensure that the stockpiles are protected from rainy and windy weather. If such conditions are anticipated, covering the piles with tarps or plastic is critical. All materials should be relatively dry at the time of proportioning and blending. This will help to achieve correct material ratios, simplify mixing, reduce clods and clay balls, and achieve homogeneity in the final mix.

14.2.6 Summary

The creation of a high-quality mix design usually requires a minimum of two material inputs. In situations where suitable soil materials are found on-site, their composition will commonly include higher amounts of silt and/or clay-sized particles than is optimum. Additionally, they may require screening or other on-site processing. Sand and gravel are normally readily available from nearby quarries or landscape supply yards and can be used to improve a high-fines site-soil. Waste and recycled, both post-consumer and post-industrial, amendments should be preferable to virgin aggregates because of the associated environmental benefits, though performance testing should be the qualifying factor. Because of the high costs associated with transporting bulk soil and aggregate materials, informed decision-making throughout the process of selecting raw materials is critical. After those stockpiled materials are assembled they should be protected from the elements.

14.3 Proportioning and mixing

The finished quality of a rammed earth wall is governed by the characteristics and proportions of its material inputs. On one hand, a completely serviceable wall may be constructed of native soil excavated from the site, placed directly into the formwork, and compacted to density. However, the future success of the rammed earth industry demands that procedures for the precise control

over quality and consistency of ingredients continue to be established and improved. To date, this control has largely existed in the form of engineering performance specifications defining acceptable ranges for: (i) input material gradations and (ii) the finished product's ultimate compressive strength. Where gradation requirements remain relatively consistent within the industry, compressive strength requirements vary by as much as a factor of ten. Upper limit compressive strengths (e.g. > 2000 psi (13.79 MPa)) are achievable only with careful formulation, high stabilization ratios and precise quality control. History and current practice in some parts of the world leave no room for debate as to the potential for success and long term serviceability of unstabilized rammed earth. With proper soil selection and compaction, such rammed earth walls can meet all realistic structural expectations for low-rise construction. However in New Zealand, Australia and North America, where much of the recent revival of rammed earth has taken place, building codes have specified compressive strengths that are unattainable without cement stabilization. For this reason, the formulations and processes to be discussed in the following sections will be related with the assumption that cement is included in the mix design.

14.3.1 Proportioning

In basic formulations, where only one raw aggregate source and one stabilizer are involved, proportioning is a simple volumetric ratio between soil and cement. A 7% cement stabilization ratio is roughly 13 parts soil to 1 part cement. Measured with shovels, it is 1 shovel of cement to 13 shovels of soil.

In larger batches, proportioning is accomplished with the buckets on front-loading tractors or skid-steer tractors. Cement is measured in sacks. One sack equals one cubic foot. Tractor buckets are typically 1/4 yard, 1/3 yard or 1/2 yard (roughly 7 cubic feet, 9 cubic feet or 13 cubic feet). The challenge when measuring with tractor buckets is to develop the ability to maintain a consistent fill line. A heaping bucket may contain as much as 50% more material (by volume) than a level bucket. When two or more equipment operators are sharing proportioning responsibilities, it's essential that they agree on and maintain consistent fill lines. Take for example, one operator using a 1/3 yard tractor bucket deposits 6 level buckets onto the mix pad, the result is roughly 2 cubic yards. For a cement ratio of 7% he would then empty 4 sacks of cement onto the pile. A second operator deposits 6 heaping buckets onto the mix pad, resulting in roughly 3 cubic yards. Four sacks of cement onto this pile would yield a stabilization ratio of 5%, a reduction in total cement content of nearly 30%.

Increased control over proportioning can be achieved using any of several volumetric mixing machines manufactured for the concrete industry. These machines are designed to give precise control over the ratio between

cement, aggregate, water and admixtures. In addition to the major equipment manufacturers such as Cementech, Elkin, Zimmerman, Airplaco and Strong, there are a few smaller companies marketing soil-blending machinery.

In formulations where two or more raw material sources are being blended, pre-proportioning materials prior to mixing is advantageous. Pre-blending the full quantity of soil and aggregate calculated for the entire wall system will yield more consistent results.

14.3.2 Mixing

As with proportioning, there are a variety of approaches to mixing materials in preparation for installation, with varying levels of precision and capital investment. Determination of the most suitable method to employ should involve deliberation on the following aspects of the project: (i) scope, (ii) budget, (iii) available time, (iv) labor skill-set and (v) desired finished wall quality. The approaches can generally be divided into two groups: ground mixing and machine mixing.

Ground mixing

This is by far the most common mixing technique (historically and in current use), and requires the least capital investment. After proportioning the soil, aggregate and cement materials on a hard, clean mixing area, the ingredients are incorporated with one another in their dry state. Once thoroughly combined, the materials are continually mixed as water is added to attain the proper moisture content. The optimum moisture content for a particular mix design is based on maximum dry density and maximum achievable compressive strength, and should be determined prior to construction through laboratory testing. Heavily affected by stabilization ratios, typically it will be in the range of 8–12% of the mix by weight.

The process of blending dry materials on a mixing pad can be accomplished in one of four ways (presented here in order of increasing capital investment): (i) by hand with shovels, (ii) with a garden type rototiller in combination with shovels, (iii) with a tractor bucket or (iv) with a rototiller mounted on a tractor used in combination with the tractor bucket. The use of a rototiller in mixing operations can yield significant improvements in mix quality, especially when dealing with soils with clods. The desired pace of the installation will influence which strategy is optimal.

With small unit set-ups, a single two- to four-yard batch can be mixed at one time using any of the above techniques, with proper moisture being maintained throughout installation either with tarps or misting. The prepared mix should be placed into the forms and compacted as quickly as is feasible, with not more than two hours elapsing between initial hydration of the cement

and installation. Special care should be taken and increased haste used in especially hot or windy conditions.

For larger form set-ups, where material requirements exceed four yards, smaller batches are preferable, with each batch mixed, moistened and delivered to the formwork as it is needed.

Machine mixing

As rammed earth projects have increased in size and complexity, structural engineering and project oversight have also increased. This is especially true in California where earthquake safety and litigation aversion are paramount. Building officials and special inspectors question the quality control when proportioning with buckets, mixing on the ground and moistening with a garden hose.

In 1990, the authors began the conversion from ground mixing to machine mixing in order to gain better control over mix consistency and directly respond to the quality control concerns of design professionals.

In machine mixing, the dry soil and aggregate are loaded into one or more stationery bins, with either a conveyor belt or a dry material auger pulling material out of the bin at a set rate and into a mixing auger. At the same time, cement is metered into the dry material. Once soil and cement are combined, the mixing auger blends the material and moves it towards the output end of the machine. Part way along the auger, water is added to the dry mix. Volumetric mixers can be adjusted to precisely control all input proportions, and to deliver the mixed material at the rate at which it is being installed in the formwork. They can be stopped and started within seconds. Small mixers are capable of producing four to five cubic yards per hour. Large mixers can produce ten or more cubic yards per hour.

The capital expense of a mixing machine will have a significant impact on operating costs for a rammed earth builder. Costs of volumetric mixers range from \$50,000 – 100,000.

There are several challenges to competitively pricing rammed earth using expensive equipment. It is important that the improvements in mixing and delivery result in better wall quality and higher strengths. Second, the forming system and crew capabilities must be such that they utilize the mixing machine's high rate of production. Third, it is advantageous if construction opportunities follow one after another to allow the rammed earth contractor to maximize equipment use thereby recouping investment costs.

14.4 Formwork

If material sourcing represents the single greatest impact on cost and environmental benefits associated with a rammed earth project, then the

choice of forming system represents the second greatest. As with cast-in-place concrete, there are numerous strategies for designing and erecting suitable formwork for rammed earth. Which of the many systems to use on any given project will be influenced by several factors: cost, structural design, reusability, desired finish and available manpower.

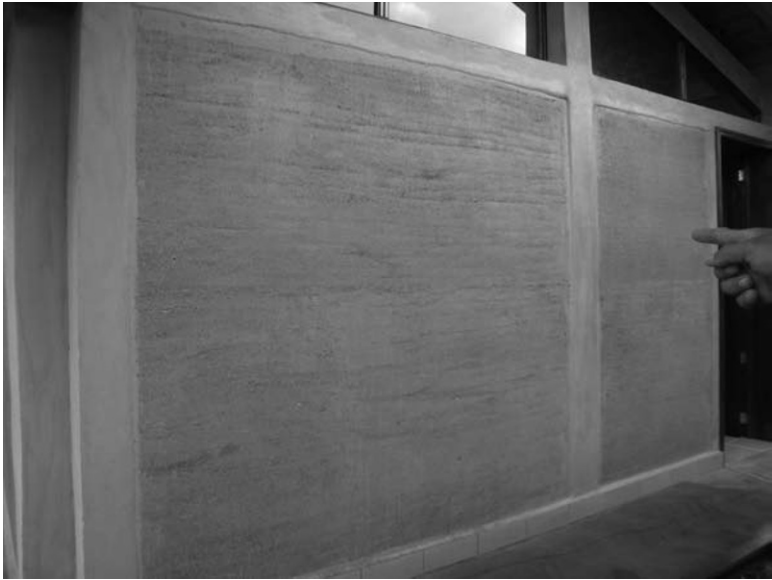
Historically, formwork for rammed earth comprised a cumbersome composite of planks and ties, assembled in place and used to define small wall units. The advantages were low cost and reusability. Disadvantages included difficult assembly, low rates of production and a lack of precision. Ragged edges and rough joint lines are inherent in this method. Relatives of this forming system are still in use today in many developing countries, where neither speed of installation nor wall surface quality are of governing importance.

In contrast, on projects where wall quality is a top priority, where panel seams and cold joints must be as inconspicuous as possible, the selection of forming systems is of significant importance. This is especially true because the rammed earth aesthetic has evolved over recent decades from a low-cost utilitarian wall system to a highly prized architectural finished wall.

14.4.1 Small panel formwork

In 1976, David Easton developed a forming system using the traditional rammed earth form as a starting point. This lightweight forming system, quick to assemble and disassemble, could be relied on to produce walls that were true and plumb, having straight tight seams. Called the California form, the system was primarily developed in response to engineering uncertainties about the earthquake safety of rammed earth construction. The early applications of the California form were to build free-standing panels of rammed earth around the building perimeter and to later cast reinforced concrete posts to lock adjacent panels together, all capped with a reinforced concrete bond beam (see Figs 14.2 and 14.3). Since that time, as test data have continued to document the strength of stabilized rammed earth, design engineers have come to rely less on the concrete frame.

The components of the California form system are: (i) plywood form faces backed by (ii) 2" × 12" walers spaced at close intervals, with (iii) full-height 'end-boards' (shutters), all held together with (iv) pipe clamps as form ties. The use of wide walers to resist pillowing of the plywood form faces horizontally allows for clamps to be spaced up to nine feet apart. The walers also provide both a ladder and a comfortable working platform at the top of the wall. The wide spacing of the clamps creates an open box form, which allows for uninterrupted and thorough compaction. The finished wall panel is uniform, dense and free of tie holes. Adjacent panels lock into place in a tongue and groove configuration after finished rammed earth panels have



14.2 Panel to panel rammed earth construction using the California form. Finished rammed earth wall with concrete posts and bond beam.



14.3 Continuous form.

been allowed to cure for at least 24 hours. Chamfer strips can be used at form panel junctures if desired.

The California form is well adapted to owner-builders and to applications in developing countries. It is inexpensive, reusable, takes up little room in storage and, being small, establishes realistic daily production milestones. Once a form is fully filled and the top layer of rammed earth is compacted, the form can be immediately disassembled and re-set. Indeed, one of the great advantages to rammed earth construction in general is that a wall or wall section does not rely on its formwork for support during curing (though performance characteristics will improve if protection from the elements is provided at this stage).

14.4.2 Continuous form systems

In contrast to the small form, panel-to-panel approach, continuous forming involves forming long sections of wall at a time (see Fig. 14.4). Broadly, there are two approaches to continuous form systems: site-built and manufactured. Advantages to continuous form systems are (i) finished walls are joint-free for longer expanses than with a panel system, (ii) daily production quotas increase and (iii) enhanced performance characteristics result from extended curing times within the formwork. Disadvantages are: (i) initial cost, (ii) potential for lumber waste and (iii) storage requirements in between uses. For single use form systems, the framing lumber and plywood can be recycled into other elements of the structure.



14.4 Installation using conveyor delivery.

Site-built forms

Site-built forms typically comprise sheets of forming plywood, wooden framing members and extractable form ties. One approach is to build $4' \times 8'$ panels of plywood and $2'' \times 4''$ s. The panels are set adjacent to one another along the foundation and $2'' \times 12''$ walers are used horizontally on opposing sides of the wall, spaced at roughly two-foot intervals. Form ties are located at roughly eight-foot intervals to pull against the walers and support the form system against expansion. If desired, additional support members, called strongbacks, can be placed vertically (perpendicular to the walers) at roughly eight-foot intervals. Strongbacks will provide added insurance that the wall will stay aligned and will allow for increased spacing of the form ties in the vertical direction. Aluminum strongbacks can be rented from several concrete supply houses or they can be fabricated of timber or double $2' \times 12'$.

A second approach to site-built formwork is to construct a frame wall along the foundation and attach the plywood sheets to the framework once it is erected, plumbed and braced. The support system of walers, ties and strongbacks is the same as described above. With a simple frame wall, plywood sheets are nailed from the face side, with nails small enough that framing lumber can be pulled away from the sheets once the wall is compacted.

For an easier disassembly, the frame wall can be built with $2'' \times 4''$ Ts. This way the plywood sheets can be attached to the frame from the backside, through the flange of the T, which allows for a cleaner disassembly. The advantages to this system over the panel system are (i) the layout of plywood seams can be adjusted to the architectural plan, (ii) no screw heads or nail heads showing on the finished wall and (iii) safe disassembly. The disadvantage to the T system is simply that it uses more lumber.

The site-built methods described above are by no means the only options, simply two with which we have had success. Carpenters and concrete form-setters will commonly feel more comfortable erecting formwork of their own design, although it should be emphasized that the forces exerted on the forms from compaction are greater than those from poured concrete.

Note that building forms for curved walls will in nearly all cases require the expertise of a knowledgeable carpenter. Depending on the radius and the symmetry of the curve, constructing, aligning, restraining and bracing a curved wall can consume many more man-hours than forming straight walls. Architects and clients should be made aware of the challenges involved and time that must be committed to building curving rammed earth walls.

Manufactured form systems

Form systems designed and manufactured for the concrete industry can be readily adapted to rammed earth construction. They are typically sold or

rented as complete packages, with supports, ties and bracing. In some cases, plywood is attached to panels, in other cases, it is to be installed on site. For large-scale projects, the supplier of rented formwork will provide take-offs and installation diagrams.

Two popular manufactured forming systems are Symons and Atlas. Symons forms are plywood panels wrapped with a steel frame. Panels are typically four feet high, held together with clips and flat ties spaced roughly two feet on center. Walls built using the Symons system are typically set and compacted to four feet in height, after which point a second level of panels is set on top of the first to allow for construction of the rest of the wall. The main advantages of the Symons system are the ease and speed of panel installation and disassembly. The disadvantages are the proximity of form ties and the creation of holes in the finished wall that will need to be patched after tie removal.

The Atlas system comprises aluminum waler beams, aluminum strongbacks and removable form ties – either taper ties or she-bolts. Forming plywood is attached to the aluminum beams on site, either once the framework is erected or on a working deck from which it can be set in place with a grade-all or crane. The advantages to the Atlas system are that wall forms are set to full wall height for one-storey structures, form ties are widely spaced to provide room for thorough and timely compaction, there are a limited number of tie holes to patch and large areas of formwork can be set and re-set quickly. Disadvantages include capital costs and the fact that equipment is required to move the forms, further adding to overall wall cost.

Both Symons and Atlas, as well as all the other manufactured systems available for rent or sale, provide special components designed to create and brace inside and outside corners.

When it comes to forming plywood: the denser the veneers, the stiffer the plywood and the straighter the wall. Quality is proportional to cost. Plywood comes in several face designations: CDX, AC, BBX, MDO and HDO and barrier film. CDX, AC and BBX will display the grain of the wood veneer. They will hold up for fewer uses and will get dinged during ramming. MDO (medium-density overlay), HDO (high-density overlay) and barrier film (high-density with a superior overlay) have the outside veneer coated with a thin film to obscure the grain. They will provide dozens of satisfactory uses. Barrier film results in the tightest finished surface.

Other systems

At least two rammed earth builders have developed unique, patented forming systems specific to their installation methods. These are: the SIREWALL form from Terra Firma Builders in British Columbia, and the Stabilform from the Associated Earthbuilding Group in Western Australia.

Permission to use the form requires that an agreement be in place with the patent holder.

14.4.3 Summary

A range of forming options are available to the rammed earth builder, with decisions regarding method of forming (continuous vs. panel-to-panel), materials selection (plywood type, manufactured vs. site-built forms), level of precision, etc. all affecting the cost of construction and the aesthetic of the final product. Furthermore, decisions made at this stage will affect production rates and other stages of wall construction, including method of material delivery and installation.

Setting and disassembling formwork takes a great deal of time (especially a continuous form that incorporates corners, door and window block-outs, and mechanical services). Depending on the forming system selected and time expended in filling the forms, the formwork phase of a project can take up to twice as many man-hours as the installation and ramming.

Two recently completed projects by the REW team confirm this calculation. Using the Atlas system to construct three 20' × 24' × 10' high modules where soil material was mixed volumetrically and delivered to the formwork by conveyor, a total of 750 man-hours was invested in form setting and 500 man-hours on compaction. (2640 total square feet of wall for a combined total of 1250 hours or roughly two square feet per man-hour labor investment.)

14.5 Installation

The process of installing the rammed earth mix into the formwork to create a finished wall can be separated into two steps: delivery and compaction. Each of these can be approached in a variety of ways. The decisions regarding which methods to use are influenced by several factors: type of forming system, site access, manpower resources, labor rates, budget, schedule and finish quality expectations.

14.5.1 Delivery

Delivery of rammed earth to the forms is a process that has seen much experimentation and innovation throughout the evolution of the industry. The most economical and historically widely used method is the bucket or basket, carried up a ladder or passed hand to hand. Innovative builders have developed a range of low-cost improvements to the hand-carried bucket, such as bottom-dumping backpacks, wheelbarrows and ramps, ropes and pulleys. These are all slow and labor-intensive methods requiring lower capital expenses. They are most appropriate to owner-builders and in developing countries.

The majority of the rammed earth industry today uses the front-loading tractor to carry material from the mixing area to the forms and a worker with a shovel standing either on top of the forms or in the tractor bucket to distribute it. A variation of the tractor bucket delivery is a hopper on a forklift. The tractor loads material on to the hopper, the forklift drives to the form and tilts the mast forward, and material is 'raked' off of the hopper and into the forms. Having a forklift in the operation frees up the tractor for additional mixing time, provides a superior working platform and facilitates delivery to taller formworks.

The REW team utilizes tracking conveyor belts to move prepared material from the mixing machine directly into the formwork in regulated course depths. Conveyors can be linked to extend filling ranges (see Figs 14.5 and 14.6).

The rate at which material can be moved from the mix area to the formwork is usually directly proportional to the capital investment in delivery equipment. Cranes and conveyors are extremely fast, but they can be expensive.

14.5.2 Compaction tools and techniques

There are two ways to compact soil: manually and mechanically. Manual compaction (by hand) is slow and laborious, although it can be every bit as effective as mechanical compaction.



14.5 Installation using conveyor delivery.



14.6 Concrete posts and bond beam yet to be poured.

Manual compaction

Historically, all rammed earth was compacted by hand, using heavy wooden rammers. Hand compaction is still the most viable method in any economy where labor is inexpensive and widely available. Steel rammers made of 1" diameter pipe and 4" square heads are an improvement over heavy wooden rammers. In all but the smallest form there will be room for extra people. Compacting in a group can remove much of the tedium from the process.

Mechanical compaction

Nearly all rammed earth builders in industrialized countries use pneumatically powered backfill tampers (see Fig. 14.7). Some experiments have been undertaken to assess the effectiveness of gas-powered whackers and electric-powered vibratory rollers, but to date the backfill tamper has proved the most versatile and effective.

Backfill tampers possess a number of attributes, the most important being: (i) weight, (ii) foot (or butt) size, (iii) strokes (blows) per minute and (iv) air pressure requirements. Manufacturers include Jet, Ingersoll-Rand and Chicago Pneumatic. Backfill tampers typically cost between \$800 and \$1200.

While large, heavy backfill tampers will yield fast and effective results, there are frequently conditions (especially where there are a large number of imbeds or prolific amounts of reinforcing steel) where less cumbersome and more maneuverable tools are preferable. In situations where even the



14.7 Pneumatic compaction at the top of a rammed earth wall.

smallest backfill tampers cannot access every area within the formwork, steel or wooden hand tampers will prove useful. Having a variety of tools available for use throughout the entire installation is ideal.

When walls are 18" or wider, it is often possible for installers to work within forms set to full wall height. In instances where wall thickness or an abundance of imbedded elements prohibit installers from getting inside the forms, installers will have to work from the top of the formwork. Small backfill tampers custom fit with long handles can be used to install walls of heights up to 12 feet, though quality control at the bottom of the wall becomes more difficult as the height of the formwork increases.

Air compressors

Pneumatic backfill tampers are powered by compressed air. The weight and total number of tampers employed on the project will dictate the required output of the air compressor, which is measured in cubic feet per minute (cfm). Where one small Jet 2T rammer will require roughly 20 cfm, a big Ingersoll Rand 441 requires nearly 40 cfm. A large nail gun compressor can marginally power a single Jet 2T. Running six Jets or two 441s will tax a small towable 100 cfm compressor. Rental yards commonly stock 185 cfm compressors, which are adequate for most installations.

The rate of soil delivery will dictate how many rammers to put into use. Delivery from tractor buckets may require one or two rammers; forklift hoppers two to four; conveyor delivery six to ten rammers.

The production quotas on the projects referenced above were 40–50 cubic yards of placed material per shift with six people on Jet 2T backfill tampers.

Strata and compaction techniques

The signature of architectural rammed earth is the horizontal lines (reminiscent of geologic strata lines) that reflect the courses (or *lifts*) of compaction. Control over the depth, density and uniformity of these lines reflects the skill of the builder. Some architects and clients prefer widely diverse line and depth, infused with different soil colors. Others specify subtle, consistent, horizontal lines. It is essential that all parties agree on the wall characteristics well before construction begins.

Soil characteristics, lift depth and lift homogeneity are the principle factors that affect the finished wall appearance. Clay soils demand more effort during compaction than sandy soils or installation using smaller lifts – depth should not exceed 7 inches to prevent the risk of under-compaction. Walls made with heavy clayey soils will have improved durability characteristics if course depths are kept down to 5 inches or less. On the other hand, sandy soils will in some cases reach satisfactory compaction at much larger lift depths (12 inches or greater). Varying lift depths in test wall panels is the best way to understand a specific material's tolerances and establish guidelines for the project.

Sharp, distinct interfaces are obtained by fully compacting each lift before adding more soil. Blended interfaces are obtained by slightly under-compacting courses and allowing the course above to integrate with the top surface of the course below.

Each course up the wall, with the exception of the top course, experiences two types of compaction: the first is the direct force of compaction from the impact of the rammer; the second is a sort of surcharge resulting from the compaction of subsequent courses on top of it. The final course at the top of the wall never benefits from the surcharge effect and, while compacting it more thoroughly is helpful, it will typically be less dense and may weather differentially. Bond beams and wall caps can alleviate this problem.

The incorporation of different colors into the rammed earth wall is a matter of both taste and technique (see Figs 14.8 and 14.9). Colors can be dramatically different from one another and used in multiple bands, or changes can be subtle and used sparingly. Coloring can be achieved naturally, with clay pigments, or with chemical dyes. Perfecting the art of the stratified earth wall is a challenge to be taken seriously and with patience.



14.8 Rammed earth wall with color changes.



14.9 Rammed earth wall with color changes.

For designs where changes in color between wall courses are intended, ground mixing necessitates the creation of separate batches on the mix pad. Distinct color strata are achieved in the wall by careful separation between batches and complete compaction of courses within the formwork. For subtle

color changes, batches are allowed to mingle and shorter wall courses may be left slightly under-compacted so that there will be some blending with subsequent courses. With careful planning and pre-construction testing, a number of approaches can be used to successfully produce a variety of aesthetics.

14.5.3 Stripping and curing

With small panel construction, forms are disassembled soon after completion (in some cases within minutes). Freshly built rammed earth walls are not yet strong. Corners can easily be broken and faces marred. Rapid drying from wind or sun will have adverse effects. Walls should be kept moist and covered with curing blankets or plastic if the full value of the stabilization and construction effort is to be realized. With stabilized rammed earth, it can be assumed that significant performance increases will continue to be realized if curing conditions are maintained for up to 28 days. Like concrete, stabilized rammed earth will be serviceable without undertaking any special effort to prevent moisture loss and exposure to the elements, but enhancement of performance characteristics is possible with the application of good curing techniques. With continuous form systems, formwork is normally left in place for several days at least, while reinforcing steel and bond beams are completed. This time can contribute significantly to the ultimate strength and durability of the finished wall.

14.5.4 Summary

Installation of a rammed earth wall consists of two steps: delivery and compaction. As in other parts of the rammed earth process, delivery can be achieved in a variety of ways, depending on the capital investment. Traditionally, delivery was by bucket or basket. Modern delivery is by tractor bucket, forklifts or conveyors – equipment to expedite installation, reduce labor requirements and, in some cases, enhance wall quality. Compaction in the forms can be achieved manually or mechanically, with mechanical compaction requiring more capital and less labor. Techniques for creating rammed earth with color striations are varied, though achieving a tasteful aesthetic takes experience and patience. Care should be taken to not damage walls during form disassembly, and they should be kept moist for a minimum of seven days.

14.6 Future trends and conclusions

How well the construction industry responds to an expanding use of rammed earth is in many regards a factor of local building practice. In regions of the

world where wood frame construction is common, solid walls – whether of stone, brick, block, concrete or rammed earth – will remain at a competitive disadvantage. The construction industries in these regions have evolved to support frame walls. Tradesmen and suppliers invest their time and resources developing skills and products to support the majority demand. On the other hand, in regions of the world where construction favoring solid wall systems predominates, rammed earth can more quickly gain recognition.

Within the US, the use of rammed earth has achieved a very small market penetration, appealing principally to clients and architects with an appreciation for the unique and compelling qualities of raw earth and of natural building systems.

What changes could help rammed earth gain traction in the construction industry? (i) expand the pool of trained installers, (ii) reduce the cost of construction, (iii) compile test data to define formulations and to document structural performance and (iv) develop a program of subsidies or incentives supporting low-carbon thermal storage wall systems.

14.7 Sources of further information

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